Theory of Commutative Algebra 14

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Lecture notes with worked examples in commutative algebra

Contents

1 Localization at a prime, exact sequences, and the module-theoretic properties of $A_\mathfrak{p}$

Let A be a commutative ring and let $\mathfrak{p} \in \mathrm{Spec}(A)$ be a prime ideal. Put

$$S = A \setminus \mathfrak{p}.$$

Then the localization of A at $\mathfrak{p}$ is

$$A_\mathfrak{p} = S^{-1}A.$$

Its elements are fractions

$$\frac{a}{s}, \qquad a \in A,\ s \notin \mathfrak{p}.$$

Definition of the natural map $A \to A_\mathfrak{p}$

There is a canonical ring homomorphism

$$\iota : A \longrightarrow A_\mathfrak{p}, \qquad a \longmapsto \frac{a}{1}.$$

Its kernel is

$$\ker(\iota) = \{a \in A : \exists s \notin \mathfrak{p} \text{ such that } sa = 0\}.$$

Hence one always has an exact sequence

$$0 \longrightarrow \ker(\iota) \longrightarrow A \xrightarrow{\iota} A_\mathfrak{p} \longrightarrow \mathrm{coker}(\iota) \longrightarrow 0.$$

If A is a domain, then ι is injective, so we get

$$0 \longrightarrow A \longrightarrow A_\mathfrak{p} \longrightarrow A_\mathfrak{p}/A \longrightarrow 0.$$

Localization of a short exact sequence

If $I \subseteq A$ is an ideal, then there is the standard short exact sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0.$$

Localization preserves exactness, so after localizing at $\mathfrak{p}$ we obtain

$$0 \longrightarrow I_\mathfrak{p} \longrightarrow A_\mathfrak{p} \longrightarrow (A/I)_\mathfrak{p} \longrightarrow 0,$$

where

$$I_\mathfrak{p} = S^{-1}I.$$

This is one of the most important exact sequences involving $A_\mathfrak{p}$.

The maximal ideal and the residue field

The ring $A_{\mathfrak{p}}$ is local. Its unique maximal ideal is

$$\mathfrak{p}A_{\mathfrak{p}}.$$

Therefore there is a natural short exact sequence

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \longrightarrow 0.$$

The quotient

$$\kappa(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$$

is called the *residue field* at $\mathfrak{p}$. Thus the sequence may be written as

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0.$$

Examples

Example 1: $A = \mathbb{Z}$, $\mathfrak{p} = (5)$. We have

$$\mathbb{Z}_{(5)} = \left\{ \frac{a}{b} \in \mathbb{Q} : 5 \nmid b \right\}.$$

The local ring sequence is

$$0 \longrightarrow 5\mathbb{Z}_{(5)} \longrightarrow \mathbb{Z}_{(5)} \longrightarrow \mathbb{F}_5 \longrightarrow 0.$$

Also the canonical map

$$\mathbb{Z} \longrightarrow \mathbb{Z}_{(5)}$$

is injective, so there is an exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}_{(5)} \longrightarrow \mathbb{Z}_{(5)}/\mathbb{Z} \longrightarrow 0.$$

Example 2: $A = k[x]$, $\mathfrak{p} = (x)$. Then

$$A_{\mathfrak{p}} = k[x]_{(x)}.$$

Taking $I = (x)$, the localized quotient sequence is

$$0 \longrightarrow (x)k[x]_{(x)} \longrightarrow k[x]_{(x)} \longrightarrow (k[x]/(x))_{(x)} \longrightarrow 0.$$

Since

$$k[x]/(x) \cong k,$$

this becomes

$$0 \longrightarrow (x)k[x]_{(x)} \longrightarrow k[x]_{(x)} \longrightarrow k \longrightarrow 0.$$

Example 3: $A = k[x, y]$, $\mathfrak{p} = (x, y)$. Then

$$A_{\mathfrak{p}} = k[x, y]_{(x,y)}.$$

Its maximal ideal is

$$(x, y)A_{\mathfrak{p}},$$

and the residue field is

$$A_{\mathfrak{p}}/(x, y)A_{\mathfrak{p}} \cong k.$$

Hence

$$0 \longrightarrow (x, y)A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow k \longrightarrow 0.$$

Example 4: $A = \mathbb{Z}/6\mathbb{Z}$, $\mathfrak{p} = (2)$. Here localization is more subtle because zero divisors are present. In this case some nonzero elements may become zero in $A_{\mathfrak{p}}$. Thus the map

$$A \longrightarrow A_{\mathfrak{p}}$$

need not be injective. This gives a concrete example where the kernel of

$$A \to A_{\mathfrak{p}}$$

is nonzero.

Abelian-group structure

As with every A-module, the localization $A_{\mathfrak{p}}$ is an abelian group under addition. Explicitly,

$$\frac{a}{s} + \frac{b}{t} = \frac{at + bs}{st}.$$

Thus, viewed only additively, $A_{\mathfrak{p}}$ is always an abelian group.

Is $A_{\mathfrak{p}}$ free as an A-module?

Usually, no. In general $A_{\mathfrak{p}}$ is *not* a free A-module.

For example, $\mathbb{Z}_{(5)}$ is not a free $\mathbb{Z}$-module. Indeed, it is not cyclic as a $\mathbb{Z}$-module, because 1 does not generate $\frac{1}{2}$, and it does not admit a basis over $\mathbb{Z}$.

So one should remember:

$$A_{\mathfrak{p}} \text{ is generally not free over } A.$$

Flatness of $A_{\mathfrak{p}}$

Localization is the basic example of a flat module. In fact,

$$A_{\mathfrak{p}} \text{ is flat as an } A\text{-module.}$$

Equivalently, for every short exact sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

tensoring with $A_{\mathfrak{p}}$ preserves exactness:

$$0 \longrightarrow M' \otimes_A A_{\mathfrak{p}} \longrightarrow M \otimes_A A_{\mathfrak{p}} \longrightarrow M'' \otimes_A A_{\mathfrak{p}} \longrightarrow 0.$$

Moreover,

$$M \otimes_A A_{\mathfrak{p}} \cong M_{\mathfrak{p}}$$

for every A-module M.

This explains why the sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

localizes to

$$0 \longrightarrow I_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow (A/I)_{\mathfrak{p}} \longrightarrow 0.$$

Faithfulness of $A_{\mathfrak{p}}$

An A-module M is called *faithful* if

$$\mathrm{Ann}_A(M) = 0.$$

For $A_{\mathfrak{p}}$, one computes

$$\mathrm{Ann}_A(A_{\mathfrak{p}}) = \{a \in A : \exists s \notin \mathfrak{p} \text{ such that } sa = 0\}.$$

Therefore $A_{\mathfrak{p}}$ is faithful if and only if the canonical map

$$A \longrightarrow A_{\mathfrak{p}}$$

is injective.

In particular, if A is a domain, then $A_{\mathfrak{p}}$ is faithful.

Faithful examples.

$$\mathbb{Z}_{(5)} \text{ is faithful as a } \mathbb{Z}\text{-module,}$$

because

$$\mathbb{Z} \longrightarrow \mathbb{Z}_{(5)}$$

is injective.

Also,

$$k[x]_{(x)} \text{ is faithful as a } k[x]\text{-module.}$$

Non-faithful example. If $A = \mathbb{Z}/6\mathbb{Z}$ and $\mathfrak{p} = (2)$, then localization kills some nonzero elements, so $A_{\mathfrak{p}}$ is not faithful as an A-module.

Summary

For a prime ideal $\mathfrak{p} \subseteq A$, the localization

$$A_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1} A$$

fits naturally into several important exact sequences:

$$0 \longrightarrow I_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow (A/I)_{\mathfrak{p}} \longrightarrow 0,$$

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0,$$

and, when $A \to A_{\mathfrak{p}}$ is injective,

$$0 \longrightarrow A \longrightarrow A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}}/A \longrightarrow 0.$$

As an A-module, $A_{\mathfrak{p}}$ is:

$$\begin{gathered} \text{abelian: always,} \\ \text{free: usually not,} \\ \text{flat: always,} \\ \text{faithful: exactly when } A \to A_{\mathfrak{p}} \text{ is injective.} \end{gathered}$$

1.1 Residue fields $\kappa(\mathfrak{p})$, stalks, and geometric meaning on $\operatorname{Spec}(A)$

Let A be a commutative ring and let

$$\mathfrak{p} \in \operatorname{Spec}(A)$$

be a prime ideal. Associated to $\mathfrak{p}$ are two fundamental local objects:

$$A_{\mathfrak{p}}, \qquad \kappa(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

The ring $A_{\mathfrak{p}}$ is the local ring of A at the point $\mathfrak{p}$, and the field $\kappa(\mathfrak{p})$ is called the *residue field* at $\mathfrak{p}$.

1. Definition of the stalk $A_{\mathfrak{p}}$

Recall that

$$A_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1} A.$$

Its elements are fractions

$$\frac{a}{s}, \qquad a \in A,\ s \notin \mathfrak{p}.$$

The ring $A_{\mathfrak{p}}$ is local, with unique maximal ideal

$$\mathfrak{p}A_{\mathfrak{p}}.$$

In algebraic geometry, if $X = \operatorname{Spec}(A)$, then the structure sheaf $\mathcal{O}_X$ satisfies

$$\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}.$$

Thus $A_{\mathfrak{p}}$ is the *stalk* of the structure sheaf at the point $\mathfrak{p}$.

2. Definition of the residue field $\kappa(\mathfrak{p})$

Since $A_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, the quotient

$$\kappa(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$$

is a field. This is the residue field at $\mathfrak{p}$.

There is a natural short exact sequence

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0.$$

Intuitively, $\kappa(\mathfrak{p})$ is obtained by taking all functions defined near the point $\mathfrak{p}$, and then identifying two such functions if they differ by something vanishing at $\mathfrak{p}$.

3. Residue field as a field of fractions

There is a very useful identification

$$\kappa(\mathfrak{p}) \cong \mathrm{Frac}(A/\mathfrak{p}).$$

Indeed, $A/\mathfrak{p}$ is an integral domain because $\mathfrak{p}$ is prime. Localizing $A/\mathfrak{p}$ at its nonzero elements gives its field of fractions, and one checks that this is exactly

$$A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

So one may think of $\kappa(\mathfrak{p})$ as the smallest field naturally attached to the point $\mathfrak{p}$.

4. Why $A_{\mathfrak{p}}$ is called a stalk

Let $X = \mathrm{Spec}(A)$. For each basic open set

$$D(f) = \{\mathfrak{q} \in \mathrm{Spec}(A) : f \notin \mathfrak{q}\},$$

the structure sheaf assigns the ring

$$\mathcal{O}_X(D(f)) = A_f.$$

The stalk at $\mathfrak{p}$ is the direct limit of all rings of sections over neighborhoods of $\mathfrak{p}$:

$$\mathcal{O}_{X,\mathfrak{p}} = \varinjlim_{\mathfrak{p}\in U} \mathcal{O}_X(U).$$

In the affine case this direct limit is canonically isomorphic to

$$A_{\mathfrak{p}}.$$

So $A_{\mathfrak{p}}$ contains the data of all regular functions defined in some neighborhood of the point $\mathfrak{p}$, where two functions are identified if they agree on a smaller neighborhood.

5. Geometric meaning of $A_{\mathfrak{p}}$

The local ring

$$A_{\mathfrak{p}}$$

captures what happens *near* the point $\mathfrak{p}$.

Globally, the ring A contains information about the whole space $\operatorname{Spec}(A)$. After passing to

$$A_{\mathfrak{p}},$$

we invert all elements not in $\mathfrak{p}$, which means that anything nonvanishing at $\mathfrak{p}$ becomes invertible. Thus only the behavior near $\mathfrak{p}$ remains visible.

This is why $A_{\mathfrak{p}}$ is the algebraic analogue of looking at germs of functions near a point in analysis or differential geometry.

6. Geometric meaning of $\kappa(\mathfrak{p})$

The residue field

$$\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$$

captures the *value* of a function at the point $\mathfrak{p}$.

If

$$\frac{a}{s} \in A_{\mathfrak{p}},$$

its image in $\kappa(\mathfrak{p})$ is the "value" of that germ at the point $\mathfrak{p}$. Since $s \notin \mathfrak{p}$, its image is nonzero in $A/\mathfrak{p}$, hence invertible in $\kappa(\mathfrak{p})$, so evaluation makes sense.

Thus the map

$$A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p})$$

plays the role of evaluating a regular function at the point $\mathfrak{p}$.

When $\kappa(\mathfrak{p}) = k$, one may think of the point as a k-rational point. When $\kappa(\mathfrak{p})$ is a larger field extension of k, the point is more general.

7. Example: $A = \mathbb{Z}$

Take

$$A = \mathbb{Z}.$$

Its prime ideals are

$$(0),\ (2),\ (3),\ (5), \ldots$$

The point $\mathfrak{p} = (5)$. Then

$$A_{\mathfrak{p}} = \mathbb{Z}_{(5)} = \left\{ \frac{a}{b} \in \mathbb{Q} : 5 \nmid b \right\}.$$

Its maximal ideal is

$$(5)\mathbb{Z}_{(5)},$$

and the residue field is

$$\kappa((5)) = \mathbb{Z}_{(5)}/5\mathbb{Z}_{(5)} \cong \mathbb{F}_5.$$

Geometrically, the point (5) is a closed point of $\mathrm{Spec}(\mathbb{Z})$, and the residue field tells us that reducing modulo this point gives values in

$$\mathbb{F}_5.$$

The point $\mathfrak{p} = (0)$. Then

$$A_{(0)} = \mathbb{Q},$$

because in a domain localization at (0) gives the field of fractions. Also

$$\kappa((0)) = \mathbb{Q}.$$

This is the generic point of $\mathrm{Spec}(\mathbb{Z})$. It is not closed, and it sees the whole space in a generic way.

8. Example: $A = k[x]$

Let

$$A = k[x].$$

The point $\mathfrak{p} = (x - a)$, where $a \in k$. Then

$$A_{\mathfrak{p}} = k[x]_{(x-a)},$$

the local ring of functions regular near $x = a$. Its maximal ideal is

$$(x - a)k[x]_{(x-a)},$$

and the residue field is

$$\kappa((x - a)) \cong k[x]/(x - a) \cong k.$$

So at a k-rational point a, the value of a regular function lies in k.

The point $\mathfrak{p} = (f)$, where $f \in k[x]$ is irreducible. Then

$$A_{\mathfrak{p}} = k[x]_{(f)},$$

and

$$\kappa((f)) \cong k[x]/(f).$$

If f has degree > 1, then $k[x]/(f)$ is a field extension of k. So the point is not necessarily k-rational, even though it is still a point of $\mathrm{Spec}(k[x])$.

9. Example: $A = k[x, y]$

Let

$$A = k[x, y].$$

The point $\mathfrak{p} = (x, y)$. Then

$$A_{\mathfrak{p}} = k[x, y]_{(x,y)},$$

which is the local ring at the origin. Its maximal ideal is

$$(x, y)A_{\mathfrak{p}},$$

and the residue field is

$$\kappa((x, y)) \cong k.$$

Geometrically, this is the point $(0, 0)$ in the affine plane.

The generic point $\mathfrak{p} = (0)$. Then

$$A_{(0)} = \mathrm{Frac}(k[x, y]) = k(x, y),$$

the field of rational functions in two variables, and

$$\kappa((0)) = k(x, y).$$

This is the generic point of the whole affine plane.

A non-closed prime ideal, such as $\mathfrak{p} = (y)$. Then

$$A_{\mathfrak{p}} = k[x, y]_{(y)},$$

and

$$\kappa((y)) \cong \mathrm{Frac}(k[x, y]/(y)) \cong \mathrm{Frac}(k[x]) \cong k(x).$$

Geometrically, (y) corresponds not to a single closed point, but to the generic point of the line

$$V(y).$$

Its residue field is the function field of that line.

10. Closed points and generic points

In $\mathrm{Spec}(A)$, not every point is closed.

Closed points. A point $\mathfrak{p}$ is closed if and only if $\mathfrak{p}$ is maximal. In that case,

$$\kappa(\mathfrak{p}) = A/\mathfrak{p}.$$

Generic points. A non-maximal prime ideal gives a non-closed point. Such a point is often a generic point of an irreducible closed subset. Its residue field is then typically a field of rational functions rather than just the base field.

For example, in

$$\mathrm{Spec}(k[x, y]),$$

the prime ideal

$$(y)$$

is the generic point of the line $y = 0$, and

$$\kappa((y)) \cong k(x).$$

11. Functions vanishing at a point

The maximal ideal

$$\mathfrak{p}A_{\mathfrak{p}}$$

consists of germs of functions vanishing at the point $\mathfrak{p}$. Passing to the quotient

$$A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$$

kills all such functions and keeps only their value at the point.

This is exactly analogous to the situation in differential geometry where germs of smooth functions at a point modulo those vanishing at the point give the ground field.

12. Summary

For a point

$$\mathfrak{p} \in \mathrm{Spec}(A),$$

the local ring

$$A_{\mathfrak{p}}$$

is the stalk of the structure sheaf at $\mathfrak{p}$, and it records the behavior of regular functions near that point.

Its maximal ideal is

$$\mathfrak{p}A_{\mathfrak{p}},$$

and the quotient

$$\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$$

is the residue field, which records the value of local functions at the point.

One has the exact sequence

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0,$$

and also the identification

$$\kappa(\mathfrak{p}) \cong \mathrm{Frac}(A/\mathfrak{p}).$$

Geometrically:

$$\begin{aligned} A_{\mathfrak{p}} &\quad \text{means “functions near the point”}, \\ \mathfrak{p}A_{\mathfrak{p}} &\quad \text{means “functions vanishing at the point”}, \\ \kappa(\mathfrak{p}) &\quad \text{means “the field of values at the point.”} \end{aligned}$$

1.2 Examples and counterexamples of free, flat, and faithful modules

In this subsection we collect concrete examples of modules that are free, flat, and faithful, together with important counterexamples. Throughout, A is a commutative ring with 1, and all modules are unital A-modules.

1. Definitions

An A-module M is called *free* if there exists a set I such that

$$M \cong \bigoplus_{i \in I} A.$$

Equivalently, M has a basis over A.

An A-module M is called *flat* if for every injective homomorphism of A-modules

$$N' \hookrightarrow N,$$

the induced map

$$N' \otimes_A M \longrightarrow N \otimes_A M$$

is injective. Equivalently, tensoring with M preserves short exact sequences.

An A-module M is called *faithful* if

$$\mathrm{Ann}_A(M) = 0.$$

That is, the only element of A acting as zero on all of M is 0.

2. First relations between the notions

Every free module is flat.

Indeed, if

$$M \cong \bigoplus_{i \in I} A,$$

then

$$N \otimes_A M \cong \bigoplus_{i \in I} (N \otimes_A A) \cong \bigoplus_{i \in I} N,$$

so tensoring with M preserves exactness.

Every nonzero free module is also faithful.

Indeed, if

$$M \cong \bigoplus_{i \in I} A$$

with $I \neq \varnothing$, and if $a \in A$ kills all of M, then in particular

$$a \cdot 1 = 0$$

in each copy of A, hence $a = 0$.

Thus:

$$\text{free} \Longrightarrow \text{flat}, \qquad \text{nonzero free} \Longrightarrow \text{faithful}.$$

The converses are false in general.

3. Free modules: basic examples

Example 1: A itself. The ring A, viewed as an A-module over itself, is free of rank 1:

$$A \cong A.$$

Example 2: finite-rank free modules. For any integer $n \geq 1$,

$$A^n = A \oplus \cdots \oplus A$$

is free of rank n.

Over $\mathbb{Z}$, this is the same as a free abelian group:

$$\mathbb{Z}^n.$$

Example 3: polynomial rings. Over

$$A = k[x],$$

the module

$$A^m = k[x]^m$$

is free of rank m.

Example 4: principal ideals generated by a non-zero-divisor. Let

$$A = k[x].$$

Then the ideal

$$(x) \subseteq k[x]$$

is free of rank 1 as an A-module, because multiplication by x gives an isomorphism

$$A \longrightarrow (x), \qquad f \longmapsto xf.$$

More generally, if $a \in A$ is not a zero-divisor and the map

$$A \longrightarrow (a), \qquad r \longmapsto ra$$

is injective and surjective, then $(a) \cong A$ as A-modules, hence (a) is free of rank 1.

4. Modules that are not free

Example 5: $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$. The module

$$\mathbb{Z}/n\mathbb{Z}$$

is not free as a $\mathbb{Z}$-module for $n > 1$, because every free abelian group is torsion-free, whereas

$$n \cdot \bar{1} = 0$$

in $\mathbb{Z}/n\mathbb{Z}$.

Example 6: $\mathbb{Q}$ over $\mathbb{Z}$. The $\mathbb{Z}$-module

$$\mathbb{Q}$$

is not free. Indeed, a free abelian group has a basis and therefore is a direct sum of copies of $\mathbb{Z}$, but $\mathbb{Q}$ is divisible:

$$\forall q \in \mathbb{Q},\ \forall n \geq 1,\ \exists r \in \mathbb{Q} \text{ with } nr = q.$$

A nonzero free abelian group is never divisible.

Example 7: localization $\mathbb{Z}_{(p)}$ over $\mathbb{Z}$. The module

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} : p \nmid b \right\}$$

is not free as a $\mathbb{Z}$-module. It is torsion-free, but not free.

A quick reason is that it is not cyclic:

$$1 \text{ does not generate } \frac{1}{q} \quad \text{for } q \neq p.$$

But it also cannot be a free abelian group of rank > 1, since it embeds into $\mathbb{Q}$, and subgroups of $\mathbb{Q}$ have very special structure. In particular, $\mathbb{Z}_{(p)}$ is a standard example of a flat module that is not free.

Example 8: residue fields are usually not free. Let

$$A = k[x], \qquad M = A/(x) \cong k.$$

As an A-module, k is not free, because it has torsion with respect to x:

$$x \cdot \overline{1} = 0.$$

So quotient modules like A/I are usually not free unless $I = 0$.

5. Flat modules: main examples

Example 9: all free modules are flat. As noted above, every module of the form

$$\bigoplus_{i \in I} A$$

is flat.

So in particular:

$$A, \quad A^n, \quad A^{(I)}$$

are flat.

Example 10: localizations are flat. For every multiplicative set $S \subseteq A$, the localization

$$S^{-1}A$$

is flat over A.

In particular, for every prime ideal $\mathfrak{p}$,

$$A_{\mathfrak{p}}$$

is flat over A.

Examples:

$$\mathbb{Z}_{(p)} \text{ is flat over } \mathbb{Z},$$
$$k[x]_{(x)} \text{ is flat over } k[x],$$
$$k[x,y]_{(x,y)} \text{ is flat over } k[x,y].$$

Example 11: $\mathbb{Q}$ is flat over $\mathbb{Z}$. Since

$$\mathbb{Q} = \mathbb{Z}_{(0)}$$

is the localization of $\mathbb{Z}$ at all nonzero integers, it is flat over $\mathbb{Z}$.

Example 12: vector spaces over a field. If $A = k$ is a field, then every k-module is a vector space, hence free, hence flat.

So over a field:

$$\text{all modules are free and flat.}$$

Example 13: over a PID, torsion-free implies flat. Over a principal ideal domain such as

$$\mathbb{Z} \quad \text{or} \quad k[x],$$

every torsion-free module is flat.

Thus:

$$\mathbb{Q}, \quad \mathbb{Z}_{(p)}, \quad (x) \subseteq k[x]$$

are flat.

This is an important source of examples of flat modules that are not free.

6. Modules that are not flat

Example 14: $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$. The module

$$\mathbb{Z}/n\mathbb{Z}$$

is not flat over $\mathbb{Z}$ for $n > 1$.

Indeed, consider the exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

Tensoring with $\mathbb{Z}/n\mathbb{Z}$ gives

$$0 \longrightarrow \mathbb{Z}/n\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/n\mathbb{Z} \longrightarrow (\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \longrightarrow 0.$$

But multiplication by n on $\mathbb{Z}/n\mathbb{Z}$ is the zero map, so the first map is not injective. Hence $\mathbb{Z}/n\mathbb{Z}$ is not flat.

Example 15: residue field k over $k[x]$. Let

$$A = k[x], \qquad M = A/(x) \cong k.$$

Consider the exact sequence

$$0 \longrightarrow A \xrightarrow{\cdot x} A \longrightarrow A/(x) \longrightarrow 0.$$

Tensoring with M gives

$$0 \longrightarrow M \xrightarrow{\cdot x} M \longrightarrow M \otimes_A A/(x) \longrightarrow 0.$$

But multiplication by x on $M = A/(x)$ is zero, so again the first map is not injective. Therefore k is not flat over $k[x]$.

Example 16: quotient modules are usually not flat. In general,

$$A/I$$

is usually not flat over A unless I is very special. The previous examples

$$\mathbb{Z}/n\mathbb{Z}, \qquad k[x]/(x)$$

show this clearly.

7. Faithful modules: main examples

Example 17: A over itself. The module A is faithful over A, because

$$\mathrm{Ann}_A(A) = 0.$$

Example 18: every nonzero free module is faithful. If

$$M \cong A^n \qquad (n \geq 1),$$

then

$$\mathrm{Ann}_A(M) = 0.$$

So

$$A^n$$

is faithful.

Example 19: $\mathbb{Q}$ over $\mathbb{Z}$. The $\mathbb{Z}$-module $\mathbb{Q}$ is faithful, because if an integer m kills all rational numbers, then in particular

$$m \cdot 1 = 0$$

in $\mathbb{Q}$, hence $m = 0$.

Thus

$$\mathrm{Ann}_{\mathbb{Z}}(\mathbb{Q}) = 0.$$

Example 20: localizations over domains. If A is a domain, then every localization $S^{-1}A$ is faithful as an A-module.

Indeed, if

$$a \in A$$

kills all of $S^{-1}A$, then in particular

$$a \cdot \frac{1}{1} = 0,$$

so

$$\frac{a}{1} = 0$$

in $S^{-1}A$. Since A is a domain, this implies $a = 0$.

Examples:

$$\mathbb{Z}_{(p)} \text{ is faithful over } \mathbb{Z},$$

$$k[x]_{(x)} \text{ is faithful over } k[x].$$

Example 21: the field of fractions of a domain. If A is a domain with field of fractions

$$K = \mathrm{Frac}(A),$$

then K is a faithful A-module.

For example:

$$\mathbb{Q} \text{ is faithful over } \mathbb{Z},$$

$$k(x) \text{ is faithful over } k[x].$$

8. Modules that are not faithful

Example 22: quotient modules A/I. For any ideal $I \subseteq A$,

$$\mathrm{Ann}_A(A/I) = I.$$

Indeed, $a \in A$ kills A/I if and only if

$$a \cdot (1 + I) = 0$$

in A/I, which means

$$a \in I.$$

Therefore A/I is faithful if and only if

$$I = 0.$$

So:

$$\mathbb{Z}/n\mathbb{Z}$$

is not faithful as a $\mathbb{Z}$-module for $n > 1$, and

$$k[x]/(x) \cong k$$

is not faithful as a $k[x]$-module.

Example 23: residue fields are not faithful in general. Let

$$A = k[x], \qquad M = A/(x) \cong k.$$

Then

$$x \cdot M = 0,$$

so

$$\mathrm{Ann}_A(M) \supseteq (x) \neq 0.$$

Hence k is not faithful as an A-module.

Example 24: a module over a product ring. Let

$$A = k \times k,$$

and consider the A-module

$$M = k \times 0.$$

Then

$$(0,1) \cdot (a,0) = (0,0)$$

for all $(a,0) \in M$, so

$$(0,1) \in \mathrm{Ann}_A(M).$$

Hence M is not faithful.

This is a useful example showing that even very natural modules over a ring need not be faithful.

9. A module can be flat and faithful but not free

This is a very important phenomenon.

Example 25: $\mathbb{Q}$ over $\mathbb{Z}$. The module

$$\mathbb{Q}$$

is flat over $\mathbb{Z}$, because it is a localization. It is faithful over $\mathbb{Z}$, because no nonzero integer kills all of $\mathbb{Q}$. But it is not free as a $\mathbb{Z}$-module.

So:

$$\mathbb{Q} \text{ is flat and faithful, but not free.}$$

Example 26: $\mathbb{Z}_{(p)}$ over $\mathbb{Z}$. Likewise,

$$\mathbb{Z}_{(p)}$$

is flat and faithful over $\mathbb{Z}$, but not free.

Example 27: $k[x]_{(x)}$ over $k[x]$. The module

$$k[x]_{(x)}$$

is flat and faithful over $k[x]$, but not free.

These examples are fundamental in commutative algebra.

10. A module can be faithful but not flat

Example 28: ideals in rings with zero divisors. Let

$$A = k[\varepsilon]/(\varepsilon^2), \qquad M = (\varepsilon).$$

Then $M \cong k$ as an abelian group, and

$$\varepsilon \cdot M = 0.$$

So M is not faithful.

Thus this example does not yet separate faithfulness from flatness.

A better source of examples comes from more complicated rings, but in ordinary introductory examples the main phenomenon is usually:

$$\text{faithful does not imply flat,}$$

although explicit elementary examples are less canonical than the standard localization examples above.

11. A module can be flat but not faithful

Example 29: localizations over rings with zero divisors. Let

$$A = \mathbb{Z}/6\mathbb{Z}, \qquad \mathfrak{p} = (2).$$

Then

$$A_{\mathfrak{p}}$$

is flat over A, because every localization is flat. But it is not faithful, because some nonzero elements of A become zero in the localization.

Thus:

$$\text{flat does not imply faithful.}$$

12. A module can be faithful but not free

Example 30: $\mathbb{Q}$ over $\mathbb{Z}$. As above,

$$\mathbb{Q}$$

is faithful but not free.

Example 31: $A_{\mathfrak{p}}$ over a domain A. If A is a domain, then $A_{\mathfrak{p}}$ is faithful, and usually not free.

So:

faithful does not imply free.

13. Summary table of examples

Module	Free	Flat	Faithful
A	yes	yes	yes
A^n	yes	yes	yes if $n \geq 1$
$\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{Z}$	no	no	no
$\mathbb{Q}$ over $\mathbb{Z}$	no	yes	yes
$\mathbb{Z}_{(p)}$ over $\mathbb{Z}$	no	yes	yes
$k[x]/(x)$ over $k[x]$	no	no	no
$k[x]_{(x)}$ over $k[x]$	no in general	yes	yes
A/I over $A,\ I \neq 0$	usually no	usually no	no

14. Final guiding picture

One should remember the following standard examples:

$$A,\ A^n \quad \text{are free, flat, and faithful;}$$

$$\mathbb{Z}/n\mathbb{Z},\ A/I \quad \text{are usually neither free nor flat nor faithful;}$$

$$\mathbb{Q},\ \mathbb{Z}_{(p)},\ A_{\mathfrak{p}} \quad \text{are standard examples of modules that are flat, often faithful, but not free.}$$

In particular, localization provides one of the most important classes of flat modules in commutative algebra, and quotient modules provide one of the most important classes of non-flat and non-faithful modules.

1.3 Concrete examples of associated graded rings $\mathrm{gr}_I(A)$

Let A be a ring and $I \subseteq A$ an ideal. Recall that

$$\mathrm{gr}_I(A) = \bigoplus_{n \geq 0} I^n/I^{n+1}.$$

Multiplication is induced from multiplication in A:

$$(a + I^{r+1})(b + I^{s+1}) = ab + I^{r+s+1}$$

for

$$a \in I^r, \qquad b \in I^s.$$

If

$$a \in I^n \setminus I^{n+1},$$

its image in

$$I^n/I^{n+1}$$

is often denoted by

$$a^*.$$

This is called the *initial form* of a.

The following examples make the construction concrete.

1. The case $A = k[x]$, $I = (x)$

Let

$$A = k[x], \qquad I = (x).$$

Then

$$I^n = (x^n)$$

for every $n \geq 0$. Hence

$$I^n/I^{n+1} = (x^n)/(x^{n+1}).$$

Each quotient

$$(x^n)/(x^{n+1})$$

is a one-dimensional k-vector space generated by the class of x^n.

Let

$$t := x + (x^2) \in I/I^2.$$

Then

$$t^n = x^n + (x^{n+1}) \in I^n/I^{n+1}.$$

So every graded piece is generated by t^n, and the multiplication behaves exactly as in a polynomial ring. Therefore

$$\mathrm{gr}_{(x)}(k[x]) \cong k[t].$$

Thus in this example:

$$\mathrm{gr}_{(x)}(k[x]) \text{ is Noetherian,}$$
$$\mathrm{gr}_{(x)}(k[x]) \text{ is an integral domain.}$$

2. The local case $A = k[x]_{(x)}$, $\mathfrak{m} = (x)$

Now let

$$A = k[x]_{(x)}, \qquad \mathfrak{m} = (x).$$

Then

$$\mathfrak{m}^n = (x^n),$$

and the same computation as above gives

$$\mathfrak{m}^n/\mathfrak{m}^{n+1} \cong k \cdot x^n.$$

Hence

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong k[t].$$

This is a concrete illustration of part (b) in the problem: since

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

is a domain, the local ring

$$A = k[x]_{(x)}$$

is a domain.

3. The case $A = \mathbb{Z}$, $I = (p)$

Let

$$A = \mathbb{Z}, \qquad I = (p),$$

where p is a prime number. Then

$$I^n = (p^n).$$

So

$$I^n/I^{n+1} = p^n\mathbb{Z}/p^{n+1}\mathbb{Z}.$$

Each graded piece is a one-dimensional vector space over

$$\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p,$$

generated by the class of p^n.

Let

$$t := p + (p^2) \in (p)/(p^2).$$

Then

$$t^n = p^n + (p^{n+1}).$$

Therefore

$$\mathrm{gr}_{(p)}(\mathbb{Z}) \cong \mathbb{F}_p[t].$$

So again:

$$\mathrm{gr}_{(p)}(\mathbb{Z}) \text{ is Noetherian,}$$
$$\mathrm{gr}_{(p)}(\mathbb{Z}) \text{ is a domain.}$$

If instead

$$A = \mathbb{Z}_{(p)}, \qquad \mathfrak{m} = (p),$$

then similarly

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong \mathbb{F}_p[t].$$

4. The case $A = k[x, y]$, $I = (x, y)$

Let

$$A = k[x, y], \qquad I = (x, y).$$

Then

$$I^n$$

consists of all polynomials with no terms of total degree $< n$. Therefore

$$I^n/I^{n+1}$$

is precisely the space of homogeneous polynomials of total degree n.

For example,

$$\begin{aligned} I/I^2 &= \langle x^*, y^* \rangle_k, \\ I^2/I^3 &= \langle (x^*)^2, x^*y^*, (y^*)^2 \rangle_k, \\ I^3/I^4 &= \langle (x^*)^3, (x^*)^2y^*, x^*(y^*)^2, (y^*)^3 \rangle_k, \end{aligned}$$

and so on.

If we introduce formal variables

$$X,\ Y$$

and send

$$X \longmapsto x^* = x + I^2, \qquad Y \longmapsto y^* = y + I^2,$$

then every homogeneous polynomial in X, Y of degree n corresponds to an element of

$$I^n/I^{n+1}.$$

This gives an isomorphism

$$\mathrm{gr}_{(x,y)}(k[x, y]) \cong k[X, Y].$$

Thus

$$\mathrm{gr}_{(x,y)}(k[x, y])$$

is a polynomial ring in two variables over k, hence is Noetherian and a domain.

5. The local two-variable case $A = k[x, y]_{(x,y)}$

Let

$$A = k[x, y]_{(x,y)}, \qquad \mathfrak{m} = (x, y).$$

The same reasoning as above shows that

$$\mathfrak{m}^n/\mathfrak{m}^{n+1}$$

is the space of homogeneous forms of total degree n, so

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong k[X, Y].$$

Thus

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

is a domain, and therefore

$$A = k[x, y]_{(x,y)}$$

is a domain, as expected.

6. The power series case $A = k[[x, y]]$, $\mathfrak{m} = (x, y)$

Let

$$A = k[[x, y]], \qquad \mathfrak{m} = (x, y).$$

Then

$$\mathfrak{m}^n$$

consists of power series all of whose monomials have total degree at least n. Hence

$$\mathfrak{m}^n/\mathfrak{m}^{n+1}$$

is again the space of homogeneous polynomials of total degree n.

Therefore

$$\mathrm{gr}_{\mathfrak{m}}(k[[x, y]]) \cong k[X, Y].$$

So the associated graded ring is Noetherian and a domain.

More generally, for

$$A = k[[x_1, \dots, x_r]], \qquad \mathfrak{m} = (x_1, \dots, x_r),$$

one has

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong k[X_1, \dots, X_r].$$

7. A non-domain example: dual numbers

Let

$$A = k[\varepsilon]/(\varepsilon^2), \qquad \mathfrak{m} = (\varepsilon).$$

Then

$$\mathfrak{m}^2 = 0.$$

So

$$\mathrm{gr}_{\mathfrak{m}}(A) = A/\mathfrak{m} \oplus \mathfrak{m}/\mathfrak{m}^2.$$

Now

$$A/\mathfrak{m} \cong k, \qquad \mathfrak{m}/\mathfrak{m}^2 \cong k\varepsilon^*.$$

Thus

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong k[T]/(T^2),$$

where

$$T = \varepsilon^*.$$

This is *not* a domain, because

$$T^2 = 0.$$

And indeed the original ring

$$A = k[\varepsilon]/(\varepsilon^2)$$

is not a domain, since

$$\varepsilon^2 = 0.$$

This is a very concrete example showing how zero divisors and nilpotents in A can appear in

$$\mathrm{gr}_{\mathfrak{m}}(A).$$

8. A reducible singular example: $A = k[[x, y]]/(xy)$

Let

$$A = k[[x, y]]/(xy), \qquad \mathfrak{m} = (x, y)/(xy).$$

Then

$$x^* = x + \mathfrak{m}^2, \qquad y^* = y + \mathfrak{m}^2$$

are nonzero elements of

$$\mathfrak{m}/\mathfrak{m}^2.$$

But since

$$xy = 0$$

already in A, we get

$$x^*y^* = 0$$

in

$$\mathrm{gr}_{\mathfrak{m}}(A).$$

In fact,

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong k[X, Y]/(XY).$$

So

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

is not a domain, and indeed

$$A$$

itself is not a domain.

9. A warning example: A may be a domain while $\mathrm{gr}_{\mathfrak{m}}(A)$ is not

The implication in part (b) of the problem is

$$\mathrm{gr}_{\mathfrak{m}}(A) \text{ is a domain } \Longrightarrow A \text{ is a domain.}$$

The converse is false in general.

Consider

$$A = k[[t^2, t^3]] \subseteq k[[t]].$$

This is a domain because it is a subring of the domain

$$k[[t]].$$

Let

$$\mathfrak{m} = (t^2, t^3).$$

Then

$$t^3 \notin \mathfrak{m}^2,$$

so

$$y^* := t^3 + \mathfrak{m}^2$$

is a nonzero element of

$$\mathfrak{m}/\mathfrak{m}^2.$$

But

$$(y^*)^2 = t^6 + \mathfrak{m}^3 = 0$$

in

$$\mathfrak{m}^2/\mathfrak{m}^3,$$

because

$$t^6 \in \mathfrak{m}^3.$$

Hence

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

has a nonzero nilpotent element, so it is not a domain.

Thus A can be a domain while

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

fails to be one.

10. General pattern in regular local rings

These examples suggest an important general pattern. If

$$(A, \mathfrak{m})$$

is a regular local ring and

$$\mathfrak{m} = (x_1, \dots, x_r),$$

then one often has

$$\mathrm{gr}_{\mathfrak{m}}(A) \cong (A/\mathfrak{m})[X_1, \dots, X_r].$$

This explains why in the standard regular examples one gets polynomial rings:

$$\begin{gathered}\mathrm{gr}_{(x)}(k[x]) \cong k[t],\\ \mathrm{gr}_{(p)}(\mathbb{Z}) \cong \mathbb{F}_p[t],\\ \mathrm{gr}_{(x,y)}(k[x,y]) \cong k[X,Y],\\ \mathrm{gr}_{(x,y)}(k[[x,y]]) \cong k[X,Y].\end{gathered}$$

So in many familiar regular situations, the associated graded ring is especially simple: it is just a polynomial ring over the residue field.

11. Summary of the main examples

Domain examples.

$$\begin{gathered}\mathrm{gr}_{(x)}(k[x]) \cong k[t],\\ \mathrm{gr}_{(p)}(\mathbb{Z}) \cong \mathbb{F}_p[t],\\ \mathrm{gr}_{(x,y)}(k[x,y]) \cong k[X,Y],\\ \mathrm{gr}_{(x,y)}(k[[x,y]]) \cong k[X,Y].\end{gathered}$$

All of these are Noetherian domains.

Non-domain examples.

$$\begin{gathered}\mathrm{gr}_{(\varepsilon)}\Big(k[\varepsilon]/(\varepsilon^2)\Big) \cong k[T]/(T^2),\\ \mathrm{gr}_{\mathfrak{m}}\Big(k[[x,y]]/(xy)\Big) \cong k[X,Y]/(XY).\end{gathered}$$

These are not domains.

Warning example.

$$A = k[[t^2,t^3]]$$

is a domain, but

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

is not a domain.

So the condition that

$$\mathrm{gr}_{\mathfrak{m}}(A)$$

be a domain is strong enough to force A to be a domain, but it is not necessary.

Comparison of $k[x]_{(x)}$ and $k[x]/(x)$

The rings

$$k[x]_{(x)} \qquad \text{and} \qquad k[x]/(x)$$

are *not* the same.

They arise from two different constructions:

1. The quotient ring $k[x]/(x)$. This is obtained by factoring out the ideal

$$(x).$$

In other words, one forces

$$x = 0.$$

Every polynomial is identified with its constant term, so there is a natural isomorphism

$$k[x]/(x) \cong k.$$

Explicitly, the quotient map is

$$k[x] \longrightarrow k[x]/(x), \qquad f(x) \longmapsto f(x) + (x),$$

and under the identification with k, this is just evaluation at $x = 0$:

$$f(x) \longmapsto f(0).$$

2. The localization $k[x]_{(x)}$. This is obtained by inverting all elements not belonging to the prime ideal

$$(x).$$

So

$$k[x]_{(x)} = S^{-1}k[x], \qquad S = k[x] \setminus (x).$$

Since a polynomial lies outside (x) exactly when its constant term is nonzero, the elements of

$$k[x]_{(x)}$$

are fractions

$$\frac{f(x)}{g(x)} \qquad \text{with} \qquad g(0) \neq 0.$$

Thus in $k[x]_{(x)}$, the element x does *not* become zero. Instead, x remains a nonunit and belongs to the maximal ideal

$$(x)k[x]_{(x)}.$$

3. The essential difference. In the quotient ring

$$k[x]/(x),$$

one has

$$x = 0.$$

In the localization

$$k[x]_{(x)},$$

one does *not* kill x. One only inverts those polynomials that do not vanish at $x = 0$.

So:

$$k[x]/(x) \cong k,$$

whereas

$$k[x]_{(x)}$$

is a larger local ring consisting of rational expressions regular at $x = 0$.

Therefore

$$k[x]_{(x)} \neq k[x]/(x).$$

4. The relation between them. Although they are not equal, they are closely related. The ring

$$k[x]_{(x)}$$

has maximal ideal

$$(x)k[x]_{(x)},$$

and its residue field is

$$k[x]_{(x)}/(x)k[x]_{(x)} \cong k.$$

Hence one has

$$k[x]_{(x)}/(x)k[x]_{(x)} \cong k[x]/(x) \cong k.$$

So the correct relationship is:

$$k[x]/(x) \cong k[x]_{(x)}/(x)k[x]_{(x)},$$

not

$$k[x]_{(x)} = k[x]/(x).$$

5. Geometric interpretation. The ring

$$k[x]/(x) \cong k$$

describes the point

$$x = 0$$

itself.

The ring

$$k[x]_{(x)}$$

describes functions defined *near* the point $x = 0$.

Thus:

$$k[x]/(x) \quad \text{corresponds to the value at the point,}$$

while

$$k[x]_{(x)} \quad \text{corresponds to germs of functions near the point.}$$

1.4 Localization in polynomial rings: concrete descriptions and examples

Let k be a field. Polynomial rings are among the best places to understand localization concretely, because one can see explicitly which denominators are allowed and what kind of geometric information the localization captures.

1. General definition

Let A be a commutative ring and let $S \subseteq A$ be a multiplicative set, that is,

$$1 \in S, \qquad s, t \in S \Longrightarrow st \in S.$$

The localization of A at S is the ring

$$S^{-1}A.$$

Its elements are fractions

$$\frac{a}{s}, \qquad a \in A,\ s \in S,$$

with the usual equivalence relation:

$$\frac{a}{s} = \frac{b}{t} \quad \Longleftrightarrow \quad \exists u \in S \text{ such that } u(at - bs) = 0.$$

The intuitive meaning is simple:

$$S^{-1}A \quad \text{is obtained from } A \text{ by forcing every } s \in S \text{ to become invertible.}$$

In polynomial rings, this means that one starts with ordinary polynomials and then allows division by selected polynomials.

2. Localization at one polynomial

Let

$$A = k[x].$$

If $f \in k[x]$, define the multiplicative set

$$S = \{1, f, f^2, f^3, \dots\}.$$

Then the localization is written

$$A_f = k[x]_f.$$

Its elements are fractions of the form

$$\frac{g(x)}{f(x)^n}, \qquad g(x) \in k[x],\ n \geq 0.$$

Example. If

$$f = x,$$

then

$$k[x]_x$$

consists of fractions

$$\frac{g(x)}{x^n}.$$

So

$$\frac{1}{x} \in k[x]_x, \qquad \frac{x^2+1}{x^3} \in k[x]_x, \qquad \frac{x+5}{x^2} \in k[x]_x.$$

But

$$\frac{1}{x+1} \notin k[x]_x,$$

because the denominator $x+1$ is not a power of x.

Thus

$$k[x]_x$$

is not the full field of fractions $k(x)$. It only allows denominators that are powers of x.

3. Localization at a prime ideal

Let $\mathfrak{p} \subseteq A$ be a prime ideal. Then

$$A_{\mathfrak{p}}$$

means localization at the multiplicative set

$$S = A \setminus \mathfrak{p}.$$

So we invert every element of A that does not lie in $\mathfrak{p}$.

This construction is especially important in algebraic geometry and commutative algebra.

4. The ring $k[x]_{(x)}$

Take

$$A = k[x], \qquad \mathfrak{p} = (x).$$

Then

$$A_{\mathfrak{p}} = k[x]_{(x)}.$$

Since a polynomial lies outside (x) exactly when its constant term is nonzero, the elements of

$$k[x]_{(x)}$$

are fractions

$$\frac{f(x)}{g(x)} \qquad \text{with} \qquad g(0) \neq 0.$$

Examples.

$$\frac{x^2+1}{x+1} \in k[x]_{(x)},$$

because

$$(x+1)(0) = 1 \neq 0.$$

$$\frac{1}{1+x} \in k[x]_{(x)},$$

because the denominator has nonzero constant term.

But

$$\frac{1}{x} \notin k[x]_{(x)},$$

because

$$x \in (x),$$

so x is not inverted in this localization.

Thus $k[x]_{(x)}$ consists of rational expressions that are regular at the point $x = 0$.

5. The ring $k[x]_{(x-a)}$

More generally, let

$$\mathfrak{p} = (x - a), \qquad a \in k.$$

Then

$$k[x]_{(x-a)}$$

consists of fractions

$$\frac{f(x)}{g(x)} \qquad \text{with} \qquad g(a) \neq 0.$$

So the denominator must not vanish at $x = a$.

Examples.

$$\frac{x^2+1}{x+1} \in k[x]_{(x-2)},$$

because

$$(x+1)(2) = 3 \neq 0.$$

But

$$\frac{1}{x-2} \notin k[x]_{(x-2)},$$

because the denominator lies in the ideal $(x - 2)$.

Geometrically, this ring is the local ring at the point $x = a$.

6. The ring $k[x]_{(0)}$

Now let

$$\mathfrak{p} = (0).$$

Since $k[x]$ is a domain, (0) is a prime ideal. Then

$$k[x]_{(0)}$$

means that we invert every nonzero polynomial, because

$$k[x] \setminus (0) = \{f(x) \in k[x] : f(x) \neq 0\}.$$

Therefore

$$k[x]_{(0)}$$

consists of all fractions

$$\frac{f(x)}{g(x)} \qquad \text{with} \qquad g(x) \neq 0.$$

But this is exactly the field of fractions of $k[x]$, usually denoted

$$k(x).$$

Hence

$$k[x]_{(0)} = k(x).$$

So:

$$k[x] \subseteq k[x]_{(x)} \subseteq k[x]_{(0)} = k(x).$$

This chain is extremely useful to keep in mind.

7. Localization in two variables

Now let

$$A = k[x, y].$$

Localization at x. The ring

$$k[x, y]_x$$

consists of fractions

$$\frac{f(x, y)}{x^n}.$$

Geometrically, this corresponds to the open set where

$$x \neq 0.$$

Localization at (x, y). The ring

$$k[x, y]_{(x,y)}$$

consists of fractions

$$\frac{f(x, y)}{g(x, y)} \qquad \text{with} \qquad g(0, 0) \neq 0.$$

Thus one may divide by any polynomial that does not vanish at the origin.

For example,

$$\frac{x^2 + y}{1 + x + y} \in k[x, y]_{(x,y)},$$

because

$$(1 + x + y)(0, 0) = 1.$$

But

$$\frac{1}{x} \notin k[x, y]_{(x,y)}, \qquad \frac{1}{y} \notin k[x, y]_{(x,y)}, \qquad \frac{1}{x + y} \notin k[x, y]_{(x,y)},$$

since all these denominators vanish at $(0,0)$.

Thus

$$k[x,y]_{(x,y)}$$

is the local ring of the affine plane at the origin.

Localization at (y). The ring

$$k[x,y]_{(y)}$$

consists of fractions whose denominator is not divisible by y. For example,

$$\frac{1}{x+1} \in k[x,y]_{(y)}, \qquad \frac{1}{x} \in k[x,y]_{(y)},$$

but

$$\frac{1}{y} \notin k[x,y]_{(y)}.$$

Geometrically, the prime ideal (y) corresponds not to a closed point, but to the generic point of the line

$$y = 0.$$

8. Comparison with quotients

It is very important not to confuse localization and quotient.

For example,

$$k[x]/(x) \cong k,$$

because in the quotient one forces

$$x = 0.$$

But

$$k[x]_{(x)}$$

does not kill x. Instead, it inverts all polynomials not divisible by x. So

$$k[x]_{(x)} \neq k[x]/(x).$$

The correct relationship is

$$k[x]_{(x)}/(x)k[x]_{(x)} \cong k[x]/(x) \cong k.$$

Thus the quotient

$$k[x]_{(x)}/(x)k[x]_{(x)}$$

is the residue field of the local ring $k[x]_{(x)}$.

9. Geometric meaning

Localization at one polynomial f corresponds to restricting to the open set where $f \neq 0$.

Localization at a maximal ideal, such as

$$(x-a) \subseteq k[x] \quad \text{or} \quad (x-a, y-b) \subseteq k[x,y],$$

corresponds to looking at functions defined near a concrete point.

Localization at a non-maximal prime, such as

$$(0) \subseteq k[x] \quad \text{or} \quad (y) \subseteq k[x,y],$$

corresponds to looking at functions near a generic point.

So localization can be interpreted as a way to focus algebraically on a chosen part of the space.

10. Summary

The most important concrete localizations in polynomial rings are:

$$k[x]_x \quad \text{allows denominators } x^n,$$

$$k[x]_{(x-a)} \quad \text{allows denominators not vanishing at } x = a,$$

$$k[x]_{(0)} = k(x) \quad \text{allows all nonzero polynomial denominators},$$

$$k[x,y]_{(x,y)} \quad \text{allows denominators not vanishing at the origin.}$$

Thus localization in polynomial rings is the algebraic mechanism for passing from global polynomials to local or partially defined rational expressions.

2 Worked exercises on membership in localizations

In the following exercises, the goal is to decide whether a given fraction belongs to a given localization. In each case, the criterion is simply: the denominator must lie in the chosen multiplicative set.

2.1 Exercises in one variable

Exercise 1

Determine whether

$$\frac{1}{x}$$

belongs to

$$k[x]_x \qquad \text{and} \qquad k[x]_{(x)}.$$

Solution.

In

$$k[x]_x,$$

the allowed denominators are powers of x. Since

$$x = x^1,$$

we have

$$\frac{1}{x} \in k[x]_x.$$

In

$$k[x]_{(x)},$$

the allowed denominators are polynomials not in (x), that is, polynomials with nonzero constant term. But

$$x \in (x),$$

so

$$\frac{1}{x} \notin k[x]_{(x)}.$$

Exercise 2

Determine whether

$$\frac{1}{x+1}$$

belongs to

$$k[x]_x \qquad \text{and} \qquad k[x]_{(x)}.$$

Solution.

In

$$k[x]_x,$$

the denominator must be a power of x. Since $x+1$ is not a power of x,

$$\frac{1}{x+1} \notin k[x]_x.$$

In

$$k[x]_{(x)},$$

the denominator is allowed if it does not lie in (x). Since

$$(x+1)(0) = 1 \neq 0,$$

we have

$$x+1 \notin (x),$$

hence

$$\frac{1}{x+1} \in k[x]_{(x)}.$$

Exercise 3

Determine whether

$$\frac{x^2+1}{x-2}$$

belongs to

$$k[x]_{(x)} \qquad \text{and} \qquad k[x]_{(x-2)}.$$

Solution.

For

$$k[x]_{(x)},$$

the denominator must satisfy

$$(x-2)(0) = -2 \neq 0.$$

So

$$x-2 \notin (x),$$

hence

$$\frac{x^2+1}{x-2} \in k[x]_{(x)}.$$

For

$$k[x]_{(x-2)},$$

the denominator must not lie in $(x-2)$. But

$$x-2 \in (x-2),$$

so

$$\frac{x^2+1}{x-2} \notin k[x]_{(x-2)}.$$

Exercise 4

Determine whether

$$\frac{x^2+1}{x^2-1}$$

belongs to

$$k[x]_{(x-1)}.$$

Solution.

The denominator is

$$x^2-1 = (x-1)(x+1).$$

Since it vanishes at $x=1$,

$$x^2-1 \in (x-1).$$

Therefore

$$\frac{x^2+1}{x^2-1} \notin k[x]_{(x-1)}.$$

Exercise 5

Determine whether

$$\frac{x^2+1}{x^2-1}$$

belongs to

$$k[x]_{(x)}.$$

Solution.

We test the denominator at $x=0$:

$$x^2-1\Big|_{x=0}=-1\neq 0.$$

So

$$x^2-1\notin (x),$$

and hence

$$\frac{x^2+1}{x^2-1}\in k[x]_{(x)}.$$

Exercise 6

Determine whether

$$\frac{x^3+x+1}{x^5-7x+2}$$

belongs to

$$k[x]_{(0)}.$$

Solution.

Since

$$k[x]_{(0)}=k(x),$$

every fraction with nonzero polynomial denominator belongs to this localization. So if

$$x^5-7x+2\neq 0$$

as a polynomial, then

$$\frac{x^3+x+1}{x^5-7x+2}\in k[x]_{(0)}.$$

Exercise 7

Determine whether

$$\frac{1}{x(x+1)}$$

belongs to

$$k[x]_x \qquad \text{and} \qquad k[x]_{(0)}.$$

Solution.

In

$$k[x]_x,$$

the denominator must be a power of x. But

$$x(x+1)$$

is not a power of x, so

$$\frac{1}{x(x+1)} \notin k[x]_x.$$

In

$$k[x]_{(0)} = k(x),$$

every nonzero polynomial denominator is allowed, so

$$\frac{1}{x(x+1)} \in k[x]_{(0)}.$$

2.2 Exercises in two variables

Exercise 8

Determine whether

$$\frac{1}{x}$$

belongs to

$$k[x,y]_{(x,y)} \qquad \text{and} \qquad k[x,y]_{(y)}.$$

Solution.

In

$$k[x,y]_{(x,y)},$$

the denominator must not vanish at $(0,0)$. But

$$x(0,0) = 0,$$

so

$$x \in (x,y),$$

and therefore

$$\frac{1}{x} \notin k[x,y]_{(x,y)}.$$

In

$$k[x,y]_{(y)},$$

the denominator is allowed if it is not divisible by y. Since

$$x \notin (y),$$

we get

$$\frac{1}{x} \in k[x,y]_{(y)}.$$

Exercise 9

Determine whether

$$\frac{1}{x+y}$$

belongs to

$$k[x,y]_{(x,y)} \qquad \text{and} \qquad k[x,y]_{(y)}.$$

Solution.

At the origin,

$$(x+y)(0,0) = 0,$$

so

$$x+y \in (x,y).$$

Hence

$$\frac{1}{x+y} \notin k[x,y]_{(x,y)}.$$

Now check whether

$$x+y \in (y).$$

This is false, because $x+y$ is not divisible by y. Therefore

$$\frac{1}{x+y} \in k[x,y]_{(y)}.$$

Exercise 10

Determine whether

$$\frac{x^2+y}{1+x+y}$$

belongs to

$$k[x,y]_{(x,y)}.$$

Solution.

We evaluate the denominator at the origin:

$$(1+x+y)(0,0) = 1.$$

So

$$1+x+y \notin (x,y).$$

Hence

$$\frac{x^2+y}{1+x+y} \in k[x,y]_{(x,y)}.$$

Exercise 11

Determine whether

$$\frac{x}{y}$$

belongs to

$$k[x,y]_{(y)} \qquad \text{and} \qquad k[x,y]_y.$$

Solution.

In

$$k[x,y]_{(y)},$$

we invert all elements not in (y). But

$$y \in (y),$$

so

$$\frac{x}{y} \notin k[x,y]_{(y)}.$$

In

$$k[x,y]_y,$$

we explicitly invert powers of y. Since the denominator is y,

$$\frac{x}{y} \in k[x,y]_y.$$

Exercise 12

Determine whether

$$\frac{1}{xy}$$

belongs to

$$k[x,y]_x, \qquad k[x,y]_y, \qquad k[x,y]_{(x,y)}.$$

Solution.

In

$$k[x,y]_x,$$

only powers of x are allowed as denominators. Since

$$xy$$

is not a power of x,

$$\frac{1}{xy} \notin k[x,y]_x.$$

Similarly,

$$\frac{1}{xy} \notin k[x,y]_y,$$

because xy is not a power of y.

In

$$k[x, y]_{(x,y)},$$

the denominator must not vanish at $(0, 0)$. But

$$xy(0, 0) = 0,$$

so

$$\frac{1}{xy} \notin k[x, y]_{(x,y)}.$$

Exercise 13

Determine whether

$$\frac{1}{1 + xy}$$

belongs to

$$k[x, y]_{(x,y)}.$$

Solution.

We compute

$$(1 + xy)(0, 0) = 1.$$

Thus

$$1 + xy \notin (x, y),$$

so

$$\frac{1}{1 + xy} \in k[x, y]_{(x,y)}.$$

Exercise 14

Determine whether

$$\frac{1}{y^2 + x}$$

belongs to

$$k[x, y]_{(y)}.$$

Solution.

To belong to

$$k[x, y]_{(y)},$$

the denominator must not lie in (y). The polynomial

$$y^2 + x$$

is not divisible by y, since it has the term x. Therefore

$$y^2 + x \notin (y),$$

and hence

$$\frac{1}{y^2 + x} \in k[x, y]_{(y)}.$$

2.3 Exercises comparing quotient and localization

Exercise 15

Is

$$\frac{1}{x}$$

an element of

$$k[x]/(x)?$$

Solution.

No. The ring

$$k[x]/(x)$$

is just

$$k.$$

In this quotient one has

$$x = 0,$$

so the symbol

$$\frac{1}{x}$$

does not make sense there. Quotients do not create fractions; they only impose relations. Thus

$$\frac{1}{x} \notin k[x]/(x).$$

Exercise 16

Compare

$$\frac{1}{x}, \qquad \frac{1}{x+1}$$

in

$$k[x]_x, \qquad k[x]_{(x)}, \qquad k[x]_{(0)}.$$

Solution.

For

$$\frac{1}{x}:$$

$$\frac{1}{x} \in k[x]_x, \qquad \frac{1}{x} \notin k[x]_{(x)}, \qquad \frac{1}{x} \in k[x]_{(0)}.$$

For

$$\frac{1}{x+1}:$$

$$\frac{1}{x+1} \notin k[x]_x, \qquad \frac{1}{x+1} \in k[x]_{(x)}, \qquad \frac{1}{x+1} \in k[x]_{(0)}.$$

This exercise clearly shows the differences among the three localizations.

2.4 A general test

The preceding exercises illustrate the basic criterion.

Exercise 17

Let

$$A = k[x], \qquad \mathfrak{p} = (x - a).$$

Show that

$$\frac{f(x)}{g(x)} \in A_{\mathfrak{p}}$$

if and only if

$$g(a) \neq 0.$$

Solution.

By definition,

$$A_{\mathfrak{p}} = S^{-1}A, \qquad S = A \setminus \mathfrak{p}.$$

So $\frac{f(x)}{g(x)}$ belongs to $A_{\mathfrak{p}}$ exactly when

$$g(x) \notin \mathfrak{p} = (x - a).$$

By the factor theorem,

$$g(x) \in (x - a) \quad \iff \quad g(a) = 0.$$

Therefore

$$g(x) \notin (x - a) \quad \iff \quad g(a) \neq 0.$$

Hence

$$\frac{f(x)}{g(x)} \in k[x]_{(x-a)} \quad \iff \quad g(a) \neq 0.$$

Exercise 18

Let

$$A = k[x, y], \qquad \mathfrak{m} = (x - a, y - b).$$

Show that

$$\frac{f(x, y)}{g(x, y)} \in A_{\mathfrak{m}}$$

if and only if

$$g(a, b) \neq 0.$$

Solution.

By definition,

$$A_{\mathfrak{m}} = S^{-1}A, \qquad S = A \setminus \mathfrak{m}.$$

Thus $\frac{f}{g}$ belongs to $A_{\mathfrak{m}}$ exactly when

$$g \notin \mathfrak{m}.$$

But

$$\mathfrak{m} = (x - a, y - b)$$

consists exactly of those polynomials vanishing at (a, b). Therefore

$$g \notin (x - a, y - b) \iff g(a, b) \neq 0.$$

So

$$\frac{f(x, y)}{g(x, y)} \in k[x, y]_{(x-a,y-b)} \iff g(a, b) \neq 0.$$

2.5 Final summary

In polynomial rings, localization is best understood by asking which denominators are allowed.

For

$$k[x]_f,$$

the denominator must be a power of f.

For

$$k[x]_{(x-a)},$$

the denominator must not vanish at $x = a$.

For

$$k[x, y]_{(x-a,y-b)},$$

the denominator must not vanish at (a, b).

For

$$k[x]_{(0)},$$

every nonzero polynomial is allowed, so one gets the field of rational functions

$$k(x).$$

The worked exercises above show that most membership questions in localizations reduce to one simple test:

Is the denominator in the multiplicative set, or not?